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MILITARY SPECIFICATION

TRACTOR, WHEELED, WAREHOUSE, ELECTRIC MOTOR DRIVEN, WALKING TYPE, 700 POUNDS DRAWBAR PULL

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers battery powered, solid rubber tired, walking type warehouse tractors. Tractors are for use in towing trailers and equipment on paved and other hard surfaces. Tractors shall be of the following types and sizes, as specified in the contract or purchase order:

Type EE - Tractor, wheeled, warehouse, battery powered, spark enclosed

Size 1 - Tractor, wheeled, warehouse, 700 pounds drawbar pull.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solocitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-52798	Enamel, Alkyd Camouflage
MIL-F-3541	Fittings, Lubrication
MIL-M-3971	Meter, Time Totalizing, Non-Hermetically Sealed, Electric, AC and DC
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-T-17479	Trailers, Platform, Warehouse, Wood

STANDARDS

Federal

FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LONSAC, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-162	Materials Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing
MS 51335	Pintle Assembly, Towing, 18,000 Pounds Capacity, Manual Release

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

AMERICAN STANDARDS

ASA C50.4	Direct - Current Generators, Direct - Current Motors, and Direct - Current Commutating Machines in General

(Application for copies should be addressed to: American Standards Association, Inc., 10 East 40th St., New York, NY 10016.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B 56.1	Safety Standard for Powered Industrial Trucks
B 56.3	Standards for Safety for Electric Battery Powered Industrial Trucks.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Standards and Recommended Practices

J 551	Standard for measurement of Electromagnetic Radiation from Motor Vehicles (20-100MHz)
J552	Recommended Practice for External Electromagnetic Suppressors
J754	Lubricant Types - Construction and Industrial Equipment

SAE Handbook

J534	Lubrication Fittings

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Standards

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(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

J925 - Minimum Access Dimensions for Construction and Industrial Machinery
J833 - Male Physical Dimensions for Construction and Industrial Equipment Design
29CFR, Chapter XVII, Part 1910
Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The tractor shall be a 700 pound drawbar pull, electric powered, equipped with a battery and electric drive motor, a solid state control, a battery discharge indicator, and as specified herein. The tractor shall be the latest model of the manufacturer's standard commercial product and shall have been in production, marketed, and in use for a minimum of one year preceding the solicitation for procurement. The introduction of normal product improvement changes in this one year period is acceptable. The tractor shall be equipped with standard instruments, components, and accessories normally required for the safe and effective operation of the tractor. The tractor shall be complete with all components that are standard with the contractor's products, whether stipulated herein or not, together with such accessories, and options, as may be specified herein. The tractor shall be equipped with all other components and parts not specifically mentioned but necessary to provide a functional machine and shall conform in quality to that normally provided to the commercial industry. Tractor requirements shall be as specified herein and in accordance with the applicable specification sheet.

3.1.1 First article. Unless otherwise specified, (see 6.2), a first article shall be submitted or made ready for approval by the contracting officer or his authorized representative.

3.2 Safety. Unless otherwise specified herein, the tractor shall conform to the requirements of ANSI B56.1, B56.3, type EE and with the Occupational Safety and Health Standards requirements of 29 CFR, Chapter XVII, Part 1910.

3.2.1 Steering. The tractor shall be so constructed that steering is accomplished by means of turning the drive wheel assembly, which shall be mounted on anti-friction bearings. The tractor shall have full 90 degrees steering to the right and left. The drive (steerable) wheel shall be turned by a tubular steel steering handle which can be conveniently operated from both the walking position in front of the tractor and from the operator's platform. The steering handle shall be hinged near the driving unit, and shall be spring loaded so that it "spring loads" up into operator's hands in the walking position. The steering handle shall automatically spring return up to a position near the operator's platform when released. The position to which the steer-

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ing handle returns when released shall be in a deadman "brake-on" position.

3.2.2 Environmental requirements. The truck shall safely operate in any ambient temperature from 0°F to +115°F.

3.2.3 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or tractor.

3.2.4 Material. All parts used in the fabrication of the machine shall be new and of the quality necessary to produce a tractor to meet the requirements described herein.

3.3 Construction. The tractor shall be complete so that it can be used for any operation for which it was designed. Construction shall be free from any characteristics or defects that will prevent the tractor from passing any of the examination(s) and tests in section 4.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such process as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of the tractor.

3.3.2 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Surfaces. All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.3.4 Painting. Unless otherwise specified (see 6.2), the tractor shall be painted in accordance with MIL-E-52798. The finish color shall correspond to 13538 of FED-STD-595.

3.3.5 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication, except where sealed permanent lubrication is provided. the types of lubricants shall be selected from, and in accordance with, SAE J754. All initial fills of lubricants shall be limited to the types approved by the component manufacturers. Grease lube fittings shall conform to SAE J534.

3.3.6 Maintainability. Provision shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system components or assembly where applicable, battery, wearing parts of lift and tilt mechanism, brakes and components, wear parts of steering assembly, tires, wheels, lights, and horn, without the removal of any major parts other than covers. If hand access openings are used on the exterior of the truck, the edges of each opening shall be smooth and the opening shall be provided with a removable or hinged cover, whenever it is required. Dimensions of access openings shall be in accordance with SAE J925 for bare hand. There shall be no interference with the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions. Drain holes with removable drain plugs shall provide for complete drainage without

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disconnection of hoses.

3.4 Components.

3.4.1 Structure.

3.4.1.1 Chassis and frame. The frame and related structure shall be fabricated from structural steel or steel castings at least equal to the size and design of manufacturer's normal commercial frame members. The frame and related structure shall be sufficiently strong and well braced to prevent racking or twisting that might result in permanent distortion or approach the elastic limit of any component part or member. The frame and related structure shall have sufficient depth to afford protection for working parts. It shall be capable of withstanding, without permanent deformation, stresses induced by capacity loads and tests in this specification, and stresses induced in formal operation. A certificate of compliance is required.

3.4.1.2 Body. The tractor body shall be constructed of heavy sheet steel or plate at least equal to that used in the manufacturer's normal commercial product, and shall enclose all parts requiring protection. It shall meet all requirements for type EE construction. Overall width shall not exceed 34 inches and overall length shall not exceed 54 inches, exclusive of pintle assembly. Length and width shall be such that the tractor can meet all the performance requirements specified herein.

3.4.1.3 Bumper plates. Tractors shall be provided with steel bumper plates or equivalent frame construction which shall provide protection for the complete outside periphery of the tractor. In addition, a bumper plate at the rear of the operator's platform shall be provided for the protection of the operator. This bumper plate shall be of solid steel plate, and shall extend upward a minimum of 10-inches. It shall extend the full width of the rear of the operator's platform. Bumper plates shall not interfere with routine servicing of major or minor components and shall be removable with ordinary hand tools if they obstruct the replacement of any component.

3.4.1.4 Operator's platform. Each tractor shall be provided with an operator's standup type platform. Platform floor shall provide a minimum front to rear clear space of 14-inches, and shall extend the full width of the tractor. The floor shall be fabricated from steel plate and shall be covered with a rubber cushion pad not less than 1/4 inch thick, cement bonded thereon. The operator's platform shall be provided with front and rear guard rails, fabricated from steel pipe or tubing, for the operator's comfort and safety. The rear guard rail shall extend between 32 and 40 inches above the platform floor. Both front and rear guard rails shall have rounded corners. The top part of the rear guard rail shall be such as to provide a comfortable back rest for the operator, or a 1-inch thick by 6-inch wide soft rubber cushion shall be provided at the top of the rear guard rail which extends the full width of the guard rail. If the cushion is provided, it shall have a durable waterproof cover which is easily cleaned. The 1-inch by 6-inch dimensions are defined as minimum dimensions.

3.4.1.5 Wheels. The tractor shall be mounted on flat-faced steel, malleable iron, or cast iron wheels, equipped with anti-friction bearings. No part of the wheels shall extend outside the plan outline of the tractor.

3.4.1.5.1 Drive wheel. Tractors shall be equipped with one drive wheel,

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with the drive motor connected to the wheel. The drive wheel shall be the front and steerable wheel of the tractor. Drive wheel tires shall be of the industrial solid rubber, cushion type with a minimum size of 10-1/2 inches outside diameter by 6-inches face width, and shall be of the pressed-on type.

3.4.1.5.2 Rear wheels. Tractors shall be equipped with two rear wheels with solid rubber tires which may be either molded-on or pressed-on. Rear wheel tires shall have a minimum size of 6-inches outside diameter by 4-inches face width.

3.4.1.6 Brakes. The brake system shall be interlocked with a switch so as to cause disconnection of the electrical circuit of the traction motor when the brake is applied, except under the conditions of dynamic braking or controlled plugging. The drive wheel shall be the only braking wheel. Brake shall be capable of stopping and holding tractor loaded with a drawbar pull of 300 pounds on a 15 percent slope.

3.4.1.6.1 Controllable braking. Efficient and convenient means shall be provided for fully controllable braking by manipulation of the handle or handle controls, from both the walking position and the riding position, by the operator.

3.4.1.6.2 Deadman brake. In addition to the braking specified in 3.4.1.6.1 above, a deadman brake shall be provided which shall be interlocked with a switch which shall interrupt the current to the traction motor when the deadman brake is actuated. A deadman brake pedal shall be located in a position on the operator's compartment floor which shall be convenient to the operator. The deadman brake shall be actuated only when both the steering handle and the deadman brake pedal are released. In addition to interrupting the current to the traction motor, the deadman brake shall cause sufficient braking force to be applied to the tractor to bring it to a stop from maximum speed within a distance of 6 feet. Deadman braking requirement on a slope shall be in accordance with 3.4.1.6.

3.4.1.7 Controls. All switch and lever type controls shall be provided with position marking (decals are not acceptable) either at the switch, or lever, or in the case of the directional speed controls, on a diagram visible to the operator. All controls shall have a directional motion insofar as possible, that is, the motion of the hand control shall be similar to the resulting motion of the tractor operation to be performed. The speed control shall be self centering so that vehicle stops when operator removes hand from control. The controls shall be operable and the instruments shall be viewable by the 5th to 95th percentile male and female, according to the dimentions in SAE J833a.

3.4.2 Electrical equipment.

3.4.2.1 Resistors. Resistors shall be of the heavy duty type rated for continuous use and shall not result in burnout of fusion during testing or normal operation.

3.4.2.2 Enclosures and shields. Controllers, contactors, resistors, terminal boards, and any other sparking devices shall be enclosed in dust tight metal atmosphere, other than through a shielded or sealed bearing, the length of the path from the inside to the outside of the enclosure shall be not less than 0.5 inch with a radial clearance of not more than 0.005 inch, or the minimum

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length of path for greater radial clearance shall be obtained from the formula L = 166C - 0.33 (where L is the length of axial path at the shaft opening in inches, and C is the radial clearance, or one-half the difference in diameter, in inches). A certificate of compliance is required.

3.4.2.3 Overcurrent devices. In addition to the requirements of 3.4.2.4 overcurrent devices shall be provided on each power circuit to protect the circuit from short circuits or prolonged overloads.

3.4.2.4 Resistor housings and heat-actuated devices. Resistor housings having horizontal surfaces which might accumulate combustible particles shall be avoided to the maximum extent practicable. Each resistor housing shall be equipped with not less than 2 thermal relays which shall cause disconnection of affected circuit and prevent the outside surface of the housing from exceeding a temperature of 228 degrees (°) Fahrenheit (F), when operating in an ambient temperature of 90°F, under any conditions of abnormal operation or malfunction of any part. Thermal relays shall be located in critical heat areas. This requirement shall be tested with the truck carrying rated load over a level concrete surface, with the controller in lowest speed for 75 percent (%) of the time until all surface temperatures have become stabilized. Resistor housings may have fins or be cooled by electro-mechanical means to aid in the dissipation of heat. While being tested, the thermal relays shall not be defeated in order to permit continuous operation. A certificate of compliance is required.

3.4.2.5 Static discharge device. Tractors shall be equipped with two replaceable electrically conductive rubber straps or non-ferrous chains to discharge static electricity through contact with the road surface.

3.4.2.6 Control panel. The control panel shall be designed for easy removal as a unit or of individual major components mounted thereon and the panel shall be readily accessible without interference from the tractor structure or other major components. Bolts, mounting studs, or nuts, which are mounted on the panel and so located that they are not readily accessible, shall be recessed in the panel and so located that they will not turn loose when the electrical assemblies are removed or replaced on the panel.

3.4.2.7 Conductors. Conductor runs shall be so located within the tractor frame and body, in conjunction with suitably designed non-combustible shielding, to provide mechanical protection from damage. All wires and cables shall be so coded by marking, color or tag for identification purposes. All wiring external to the tractor body and frame, which is subject to damage by moving components of the tractor or when maneuvering the tractor, shall be encased in rigid metal conduit, except that flexible metal conduit may be used at flexure areas; similar protection shall be provided for exposed wiring runs in other areas. Conductors passing through compartment walls or other metal barriers shall be protected by loom, rubber hose, or equal, and shall be securely anchored to prevent movement of conductors relative to the opening. All terminal connectors shall be fastened with spring lock washers, lock nuts, or other means of preventing accidental loosening as a result of vibration or unintentional movement of wired terminal ends. All conductors emerging from enclosure(s), except battery leads, shall be through fittings, so that sparks shall not propagate to the atmosphere and dust not enter from the atmosphere into the enclosure(s).

3.4.3 Propulsion.

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3.4.3.1 Motor and motor accessories.

3.4.3.1.1 Motors. Traction motors shall be a series of wound, heavy duty industrial type, with a high starting torque, designed for efficient operation, depending on the applied voltage. Motors shall be capable of withstanding the stresses and current loads induced by normal operation and by the tests required herein. The resultant strains and temperature rise shall not cause failure of any component part, excessive pitting or other than normal blacken ing of the commutators without external cooling. Motors shall be capable of withstanding sustained operation without evidence of insulation breakdown, short circuit or failure of any component as a result there from. Motors shall be provided with one or more thermal relay switches which shall cause disconnection of affected motor(s) as necessary to prevent the outside surface temperatures of the motor(s) from ever exceeding 228°F. Thermal relays shall be located in critical heat areas and shall interrupt the current only to the overheated motor(s). Motor enclosures shall be of the totally enclosed type. Armature conductors shall be brazed to the commutator risers. The armature commutator bars shall be of silver copper bearing material having a softening temperature of approximately 700°F. Motor lubricant shall be of a suitable type for the tests set forth herein and for normal operation. A certificate of compliance is required.

3.4.3.2 Insulation. Insulation for motors shall be either Class B, F or H or combination thereof, as defined in ASA Standard C50.4. Insulation shall be selected to give the best balance of over-all operating characteristics of torque, speed, and temperature rise. The class of insulation shall be determined by the material with the lowest permissible temperature rise used in the insulation system. Armature and fields need not have the same class of insulation, but each part shall meet the temperature rise of its class of insulation. The electrical and mechanical properties of the insulated winding shall not be impaired by the application of the temperature permitted for the insulating material.

3.4.3.3 Drive unit. The drive unit shall be of the double reduction gear type and shall operate in a constant oil bath. The complete gear train shall be mounted on ball or roller bearings. With drive wheel locked, the drive assembly shall be capable of withstanding a static test load applied to the motor shaft equal to 150 percent of stall torque developed by the drive motor using specified batteries, without failure, fracture, or deformation of any component. A certificate of compliance is required.

3.4.3.4 Controller. The speed controller shall provide for stepless control over the entire speed range, zero to maximum, in both forward and reverse directions. Speed and directional controls shall be mounted on the walking (steering) handle. Controls shall automatically return to neutral upon release. The controller shall provide means to permit precision maneuvering in restricted areas. Automatic means shall be provided to prevent reversal of current flow to the motor when the tractor speed is above 1/4 mile per hour. Contractors shall be designed for the service for which used and shall have easily replaceable tips of copper alloy or silver alloy composition. There shall be no excessive pitting or fusion of contactors, nor any other failure or malfunction of the controller, during testing specified herein or in normal use. All contactors shall be easily and readily accessible.

3.4.3.5 Battery and battery accessories.

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3.4.3.5.1 Battery. Battery/batteries shall be furnished with the tractor by the contractor. Two batteries may be furnished with each tractor. Batteries shall be shipped in the tractors charged and wet with electrolyte. A cover shall be supplied with the battery. The battery cover shall be so constructed so as to permit routine inspection and servicing of the battery, such as checking the water level, without removing the battery from the tractor.

3.4.3.5.2 Battery leads and connectors. Battery leads shall be American Wire Gauge, size number 0 or large rubber covered cables. Connectors shall be two-contact type Yale K9013 or equal. Other types of conectors will be permitted if adaptors are supplied with each battery that will permit ready connection of the batteries to chargers having two-contact type Yale K9013 connectors. When connectors are mounted outside the battery compartment, provision shall be made for locking each connector with a padlock to be furnished by the Government at destination. Charging connectors shall be so arranged that movement of the tractor cannot be accomplished while a battery is connected to the charging source. Battery leads shall be as short as possible to minimize resistance losses.

3.4.3.5.3 Battery tray. The tractor battery shall be placed in a battery tray which shall be so constructed as to permit easy removal from the tractor. The inside of the tray shall be finished with an acid resistant paint to resist deterioration through normal contact with the electrolyte. The tray shall be easily accessible for cleaning and battery servicing and shall contain no inaccessible pockets or reservoirs which are difficult to keep clean. Effective means shall be provided for securing the battery in the tray so that battery movement during tractor travel is prevented.

3.4.4 Accessories.

3.4.4.1 Hour meter. An electrical operation hour meter which registers the number of traction motor operating hours shall be provided. It shall be flush mounted in a protected and visible position in the operator's compartment, and shall register only the number of traction motor operating hours. The hour meter shall be in accordance with the requirements of MIL-M-3971, type 1.

3.4.4.2 Key-operated switch. Each tractor shall be provided with a key-operated "On-Off" switch which shall allow the tractor to be operated only when the switch is in the "On" position. The key shall be removable from the switch only when the switch is in the "Off" position.

3.4.4.3 Horn. The tractor shall be equipped with an electric horn which shall be actuated by a pushbutton provided and mounted for convenient operation by the operator from either the walking or the riding position. The horn shall be operable without requiring the operator to release the steering or directional controls. Horn button assembly and electrical wiring for the horn shall be moisture and weather resistant.

3.4.4.4 Pintle assembly. The rear of the tractor shall be equipped with a pintle assembly capable of efficiently and conveniently connecting with trailers or equipment having couplers fabricated in accordance with the requirements of MIL-T-17479. The pintle assembly shall be located at a mean nominal engagement height of 13 inches above ground level, with provisions for fixed height adjustment to 12 and 14 inches. Pintle assembly and mounting shall be capable of withstanding, without permanent deformation, stresses induced by capacity loads and tests required herein or in normal

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operation. The pintle assembly shall conform to MS 51335, part number MS 51335-2. A certificate of compliance is required.

3.5 Performance.

3.5.1 Ultimate drawbar pull. The ultimate drawbar pull at 13-inch coupler height shall be not less than 700 pounds when tested in accordance with the requirements of section 4 herein.

3.5.2 Sustained drawbar pull. The sustained drawbar pull at 13-inch coupler height shall be not less than 200 pounds when tested in accordance with the requirements of section 4 herein.

3.5.3 Right angle turn. Tractors shall be capable of turning in 90 degree intersecting aisles of equal widths of 46-inches each.

3.5.4 Speed. While towing or pushing rated loads (200 pounds sustained drawbar pull), tractors shall be capable of a forward speed and a reverse speed of at least 4.0 miles per hour. Under "no load" conditions except for a 175 pound operator riding tractor, tractors shall be capable of a forward speed and a reverse speed of at least 5.0 miles per hour. Speed requirements are to be tested as specified in section 4 herein.

3.6 Other requirements.

3.6.1 Bearings. Rotating parts shall be mounted on ball or roller bearings to the extent normally provided in standard commercial practice and in accordance with the best automotive practice. Bearing load capacity, including applicable safety and life factors, as recommended by the bearing manufacturer shall be equal to or greater than maximum bearing loading under normal truck operation. A certificate of compliance is required.

3.6.2 Lubrication. Readily accessible means shall be provided for applying and, where necessary, checking the level of lubricant in all parts and components. Oil drain plugs and connections shall provide for ready drainage using ordinary maintenance shop tools and shall be so located as to provide ready drainage with no interference from frame members or other parts of the tractor structure. Drainage shall not require the use of flexible connections or other special devices.

3.6.3 Grease fittings. Grease fittings shall be provided where required in accordance with good automotive practice, and shall comply with MIL-F-3541.

3.6.4 Protection from weather and dust. All surfaces and parts which could ordinarily be damaged by dusts or liquids, including rain, snow, liquid runoff, and splash from shallow puddles, shall be suitably enclosed, baffled, shielded or sealed to prevent such damage.

3.6.5 Radio interference suppression. Tractor shall be sufficiently free of radio interference so that it can pass the applicable requirements of SAE J551 and J552 (see 4.4.7).

3.7 Nameplate. Unless otherwise specified (see 6.2), a corrosion resistant metal nameplate permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each tractor.

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FSN
Manufacturer's Name
Manufacturer's Model Number
Manufacturer's Serial Number
Contract Number
Date of Manufacture
Gross Vehicle Weight
Draw Bar Pull
US

3.7.1 Registration data. The contractor shall furnish registration data as specified (see 6.2).

3.8 Handbooks of instructions and parts list. Unless otherwise specified (see 6.2), two copies each of the manufacturer's standard operators manual and parts list shall be plated in a waterproof envelope marked "Technical Data" and securely attached to each tractor.

3.9 Workmanship. Workmanship of the tractors and accessories shall be of the quality prevailing among manufacturers normally producing tractors of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Preproduction inspection.

b. Quality conformance inspection.

4.2.1 Preproduction inspection. Preproduction inspection shall be applied to the preproduction sample. Failure of the preproduction sample to pass the examination or one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), preproduction inspection shall consist of the examination in 4.3 and all tests in 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Failure of the machine to pass the examination or one or more tests shall be cause for rejection. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the following:

a. Examination (4.3)

b. Functional test (4.4.8)

c. Inspection of preservation, packaging, packing and marking for shipment and storage (4.5).

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4.3 Examination. The tractor shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests.

4.4.1 Ultimate drawbar pull. The tractor pintle assembly shall be connected to an anchor of sufficient strength, using a steel cable or chain, with a Dillon dynamometer or equivalent connected between the pintle assembly and the anchor. The tractor shall be stationed on level brushed concrete for this test. No friction increasing compound shall be used. The tractor shall be started forward and slowly accelerated until the dynamometer gauge reads 700 pounds. The pintle assembly shall be attached to the tractor at a height of 13 inches, plus or minus 1/4 inch above the floor, and the cable or chain shall be secured to the anchor at a height of 13 inches plus or minus 1/4 inch above the floor, to perform the above test. The operator shall ride in the operator's compartment to perform this test, and the combined weight of the operator and added weights in the operator's compartment shall be 175 pounds. Failure to pass the test shall be cause for rejection.

4.4.2 Sustained drawbar pull. With the tractor and trailer on level brushed concrete, and with the tractor pintle assembly attached to the tractor at a height of 13 inches plus or minus 1/4 inch above the floor, a Dillon dynamometer or equivalent shall be attached, with the aid of steel cable or chain, to the trailer attachment, which shall be at a height of 13 inches plus or minus 1/4 inch above the floor, the trailer shall then be loaded until, when the tractor pulls the trailer, the dynamometer reading is 200 pounds. The dynamometer shall then be disengaged and the tractor shall be attached directly to the loaded trailer by the pintle assembly. The dynamometer and cable or chain, and all other material used to attach the dynamometer to the tractor and trailer, shall be weighed, and an equal weight shall be added to the load on the trailer. The tractor shall then pull the loaded trailer over level brushed concrete for a period of 20 minutes at 44 feet per minute (Plus or minus 10 feet per minute) and 20 minutes at maximum speed, without stopping during the 40 minute test period. The operator shall ride in the operator's compartment to perform the above test, and the combined weight of operator and added weights in the operator's compartment shall be 175 pounds. Failure or malfunction of any component from this test shall be cause for rejection.

4.4.3 Right angle turn. Ninety degree intersecting aisles of equal widths of 46-inches each shall be set up using movable partitions at least 4-feet high. The tractor shall perform the right angle turn at least once in not more than 3 trials without reversal of direction and without touching the partitions. The test shall be performed in both the forward and the reverse directions. Failure to pass the test in both directions (it must pass in each) shall be cause for rejection.

4.4.4 No-load speed. This test shall be conducted on a 188 foot paved level course. The operator shall ride in the operator's compartment to perform the test, and the combined weight of the operator and added weights in the operator's compartment shall be 175 pounds. There shall be no additional load on the tractor for this test. The tractor shall be accelerated through an approach distance, timed through an 88 foot test length, and then brought to a stop. Three runs shall be made forward in highest speed, and three runs shall be made in a reverse direction in highest speed. The time for the three 88

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foot runs in a forward direction shall be added and divided by 3 to obtain the average time per run. The time for the three 88 foot runs in a reverse direction shall be added and divided by 3 to obtain the average time per run. The average speed in miles per hour in the forward direction shall be obtained by dividing 60 by the average time in seconds. The average speed in the reverse direction shall be obtained in the same manner. The average forward speed shall be at least 5.0 miles per hour, and the average reverse speed shall be at least 5.0 miles per hour. Failure to pass this test shall be cause for rejection.

4.4.5 Full load speed. This test shall be conducted identically to the test in paragraph 4.4.4 above, except that the tractor shall tow a trailer having 200 pounds drawbar pull. The drawbar pull shall be determined in the same manner as in 4.4.2. The average forward speed shall be at least 4.0 miles per hour. The average reverse speed (determined with the tractor pushing the trailer in reverse, still connected to the pintle assembly) shall be at least 4.0 miles per hour. Failure to pass this test shall be cause for rejection.

4.4.6 Preproduction post-test inspection. Preproduction post-test inspection shall be performed after the preproduction testing is completed. The post-test inspection shall include all inspection specified in paragraph 4.3 herein for acceptance examination. Particular attention shall be paid to the proper functioning of brakes, speed and directional controls, steering, horn and hour meter, and structural or other deformations, defects, or malfunctions. In addition, the commutator of the drive motor shall be examined for excessive pitting or blackening. Contactors and resistors shall also be examined for signs of excessive deterioration. Deformation, malfunction, or excessive deterioration of any component of the tractor shall be cause for rejection.

4.4.7 Radio interference suppression test. Tractor shall be tested for radio interference in accordance with SAE J551 and J552.

4.4.8 Functional test. Each truck shall be driven a sufficient distance (50 feet minimum) so that proper operation of brakes, traction motor and steering can be verified.

4.4.9 Braking test. The preproduction model, operated over the test course specified in 4.4.2, shall be tested for compliance with braking distance requirement of 3.4.1.6.2. Braking capability on a 15 percent slope shall be tested with tractor loaded to 300 pounds drawbar pull. The test shall be applied twice, once with the tractor traveling in each direction. The tractor speed down the slope shall not be less than 88 feet per minute when the brake is applied. The brake shall hold the loaded tractor on the slope for not less than 30 minutes without slippage. The above test shall be repeated using the deadman brake.

4.5 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in section 5.

5. PACKAGING

5.1 Preservation and packaging shall be level A or C as specified in the invitation for bids or contract (see 6.2).

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5.1.1 Level A. Preservation, packaging and packing shall be in accordance with MIL-STD-162.

5.1.2 Level C. Preservation, packaging, and packing shall be in accordance with commercial practice.

5.2 Marking. In addition to any special marking required, marking shall be in accordance with MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use. Tractors described herein are intended for use in towing trailers and other equipment over pavement or other hard surfaces, both indoors and outdoors, except under extremely adverse weather conditions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. First article requirement (see 3.1.1)
c. Painting, if different (see 3.3.4)
d. Nameplate, if different (see 3.7)
e. Type of registration data required (see 3.7.1)
f. Handbook of instructions and parts lists, if different (see 3.8)
g. Preproduction inspection, if different (see 4.2.1)
h. Quality conformance inspection, if different (see 4.2.2)
i. Preservation, packaging, and packing, if different (see 5.1)
j. Level of preservation, packaging, and level of packing required (see 5.1)
k. Marking, if different (see 5.2).

6.3 Preproduction sample. The sample may be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. The preproduction sample may be shipped as part of the contract. Approval of a preproduction sample shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

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